

   GLEBSA CONSORTIUM INSPECTION AND TEST PLAN STEEL RAISED FLOOR INSTALLATION 									
Project : Green Line Extension Mo Chit - Saphan Mai - Khu Kot Owner : BMA/KT Contractor : BTSC ITP no. : xxxxxx Page : 1/1									
No.	Activity	Stage/Frequency	Acceptance		Specification/ Ref. Document	Inspection and Test Record	Inspection Category		Remarks
			Method	Criteria			Contractor	MESCGLN	
1	Document Review								
	Review the status of documents and drawings required for steel raised floor installation.	Before installation	Review	No-objection or no-objection with comments	Approved letters	ITR 015-011-R2	H	W	
2	Incoming Inspection								
	Verify the quality and quantity of materials and supplies for steel raised floor received on site	Each delivery	Review Visual inspection Measure	Conform to Material Approval/Drawings No defects Dimension and size	Material approval Installation drawing	ITR 015-011-R2	H	W	
	Check HDG Thickness of steel members	Each delivery	Measure	> 50um	Material approval	ITR 015-011-R2	H	W	
3	Test Equipment Inspection								
3.1	Checks the calibration status of measuring test equipment and tools	Before testing and inspection	Visual inspection	Valid calibrated Good Condition	Calibration certificate Calibration label	ITR 015-011-R2	H	W	
4	In-process Inspection								
4.1	Setting out and location	During construction	Measure	Conform to drawings	Installation drawing	ITR 015-011-R2	S	S	
4.2	Foundation inspection	During construction	Visual inspection and measure	Conform to drawings	Installation drawing	ITR 015-011-R2			
4.3	Steel members inspection	During construction	Visual inspection and observation	Conform to drawings	Installation drawing	ITR 015-011-R2	S	S	
4.4	Grouting inspection	During construction	Visual inspection and observation	Conform to drawings	Material approval	ITR 015-011-R2	S	S	
4.5	Earthing and Bonding inspection	During construction	Visual inspection	Visual inspection	Visual inspection	ITR 015-011-R2	S	S	
5	Final Inspection								
	Check conformance of the finished steel raised floor	Before handover	Visual inspection	Visual inspection	Visual inspection	ITR 015-011-R2	H	W	
	All completed Works clean and tidy	Before handover	Visual inspection	Visual inspection	Visual inspection	ITR 015-011-R2	H	W	
H - Hold Point: A point beyond which work may not proceed without the written confirmation of ER W - Witness Point: A point provides Contractor or ER with the opportunity to witness the inspection or test of the works. S - Surveillance: Interim monitoring of any stage of the work in progress by Contractor									

TEST & INSPECTION RECORD

STEEL RAISED FLOOR INSTALLATION

Project : Green Line Extension Mo Chit - Saphan Mai - Khu Kot

Owner : BMA/KT

ITR no.: 015-011-r2

Location :

Contractor : BTSC

No	Description	Acceptance Criteria	Cat.	Acceptance		Inspected	Witness
				Yes	No		
1	Document Review						
	Material submission	All documents have no-objection or no-objection with comments	H				
	Method statement		H				
	Installation drawing		H				
2	Incoming Inspection						
2	Quality and workmanship of materials						
	No defects and imperfections	Material approval	H				
	New and unused		H				
	Grade in conformity Spec/Contract		H				
2	Foundation						
	Adhesive anchoring systems	Conform to Material approval	H				
	Stud bolts&nuts (Type and size)		H				
	Cement grout		H				
2	Steel members						
	Check dimensions as per drawings	Installation drawing	H				
	Check thickness of HDG/Manufacture certificate	> 50 μm	H				
2	Earthing and bonding						
	Check earth wire size	Material approval	H				
	Check earth terminal bolt and nut sizes		H				
3	Test Equipment						
3	Galvanized thickness meter	Valid calibration	H				
3	Automatic level	Valid calibration	H				
4	In-Process Inspection						
4	On-site check positions, levels and dimensions as per drawing	Installation drawing	S				-
4	Foundation						
	a) Location of anchor bolts	Installation drawing	S				-
	b) Check elevation of foundation	Installation drawing	S				-
	c) Depth of anchor bolts	Installation drawing	S				-
	d) Confirm vertically columns aligned	Installation drawing	S				-
	e) Check nuts tightening	Method statement	S				-
4	Steel members						
	a) All steel members correctly installed	Installation drawing	S				-
	b) Correctly install handrail, stairs and floor grating	Installation drawing	S				-
	c) Check alignment in all directions	Installation drawing	S				-
	d) All bolts, nuts and washers are correctly mounted	Installation drawing	S				-
	e) Correctly tightening for all bolts.	Method statement	S				-
4	Grouting of base plates						
	a) Check formworks are firmly in place and watertight	Installation drawing	S				-
	b) Mix cement grout as manufacture instruction	Method statement	S				-
	c) Pour cement, compact completely remove air voids	Method statement	S				-
	d) Check surface of cement after remove formwork	Method statement	S				-
5	Earthing and bonding						
	a) Installed as per drawings and specifications	Installation drawing	S				-
	b) All bolts and nuts are tighten securely	Method statement	S				-

TEST & INSPECTION RECORD

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5 Final Inspection						
	Final visual & dimensional check	Installation drawing	H			
	Verify elevation of steel raised floor with Auto level	700+/-10 mm	H			
	Verify earth bonding have been installed correctly	Installation drawing	H			
	Verify foundation grouting surfaces	No cracked	H			
	Check for cleanliness	Clean & tidy	H			
NOTES: Cat: Inspection Category H - Hold Point; A point beyond which work may not proceed without the written confirmation of ER. S - Surveillance; Interim monitor any stage of the work in progress by Contractor.			FINAL INSPECTION RESULT: _____ Accepted _____ Accepted with comment _____ Not accepted			
Comments:						
The Installation of the above Works have been satisfactorily completed in accordance with the Drawings & Specifications						
Responsibility	Company	Name	Position	Signature	Date	
Inspected by	AMR					
Verified by	BSA					
Verified by	BTSC					
Witnessed by	MESGLN					

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GLE-BSA CONSORTIUM

**BANGKOK MASS TRANSIT SYSTEM GREEN LINE EXTENSIONS PROJECT****Bearing to Samut Prakarn & Mo Chit to Khu Khot****Quality Control Check Sheet for Steel Raise Floor Installation**

Main Type : _____

Date : _____

Sub.Type/Element : _____

Sta.-Sta./Floor : _____

Location/Area : _____

Sub.Contractors : _____

EN = SITE ENGINEER , FM = FOREMAN , LAB = LABORATORY , EE = ELECTRICAL

ER = EMPLOYER'S REPRESENTATIVE , SU = SURVEY OR QUALITY CONTROL PERSON , DWG.NO. = DRAWING NUMBER

ITEM	DESCRIPTION	ACCEPTANCE CRITERIA	ACCEPTANCE			INSPECTED BY			Remark
			YES	NO	N/A	Contractor	BTSC	Consult	
During-Construction Checks									
A	Material								
1	Material ; Ref Doc.No.	FM/EN	Material Approval						
	Check Item of Main materials								
	☐ Steel support (column) w/Hot Dipped Galvanized	FM/EN	Material Approval						
	☐ Steel support (beam) w/Hot Dipped Galvanized	FM/EN	Material Approval						
	☐ Grating w/Hot Dipped Galvanized	FM/EN	Material Approval						
	☐ Stair w/Hot Dipped Galvanized	FM/EN	Material Approval						
	☐ Hand rail w/Hot Dipped Galvanized	FM/EN	Material Approval						
	☐ Epoxy adhesive ☐ Stud ☐ Accessories	FM/EN	Material Approval						
B	In-Process Inspection								
2	Preparation								
2.1	Alignment and Perpendicularity	FM/EN	Installation Drawing						
2.2	Checking Condition, level and cleanliness sub floor surface	FM/EN	Method Statement						
2.3	Checking type and dimension of steel raise floor system	FM/EN	Installation Drawing						
2.4	Checking the others under raise floor no obstruct	FM/EN	Method Statement						
3	Installation								
3.1	Set and marking the starting point and control line	FM/EN	Installation Drawing						
3.2	Install steel support (column)	FM/EN	Installation Drawing						
3.3	Checking and adjust alignment and level of steel support	FM/EN	Installation Drawing						
3.4	Install steel beam support	FM/EN	Installation Drawing						
3.5	Checking and adjust alignment and level of steel beam	FM/EN	Installation Drawing						
3.6	Checking nut fastening overall	FM/EN	Installation Drawing						
3.7	Grouting cement under plate of steel support	FM/EN	Installation Drawing						
3.8	Checking surface of cement grouting	FM/EN	Installation Drawing						
3.9	Laying steel grating on structure system	FM/EN	Installation Drawing						
3.10	Install accessories	FM/EN	Installation Drawing						
4	Finished Installation								
4.1	verify of safely	FM/EN	Installation Drawing						
	- Condition of Equipment and Materials	FM/EN	Installation Drawing						
	- Location and Position of Equipment	FM/EN	Installation Drawing						
Post-Construction Checks									
C	Pre-handover to Next Works (If Applicable)		Completion of Scope						
5	Finishes works with no defect	FM/EN	Visual Check						
6	Cleanliness, maintain conditions and final protection works	FM/EN	Visual Check						
D	Final Inspection								
7	Request ER' for inspection	FM/EN	Submittal Document						

COMMENT :**Recorded by : (BSA-AMR / Foreman)**

Signature : _____ Date : _____

Name : _____

Comments (Items to be rectified/amended)☐ No objection ☐ Not Acceptable, please resubmit

Action by : (BSA-AMR / Engineer)

Signature : _____ Date : _____

Name : _____

Witness By : (Employer's Representative)

Signature : _____ Date : _____

Name : _____